Caifen Liu

PhD Candidate in Epidemiology and Biostatistics

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EDUCATION

The University of Hong Kong Hong Kong SAR, China
PhD in Epidemiology and Biostatistics Sep 2022–Sep 2026 (expected)
Thesis: *Estimation and Interpretation of COVID-19 Severity and Vaccine Effectiveness*.
Supervisor: Prof. Ben Cowling.

Southern University of Science and Technology Shenzhen, China
MSc in Statistics Sep 2018–Jun 2020
Thesis: *Bivariate Binomial Distribution for Comparing Binomial Observations in Two Correlated Groups*.
Supervisor: Prof. Guo-Liang Tian.

Hunan University Changsha, China
BE in Materials Science and Engineering Sep 2014–Jun 2018

RESEARCH INTERESTS

I am interested in developing and applying causal inference methods to observational health data to reduce bias, strengthen the validity of causal conclusions, and support clinical decision-making.

Methodology: causal inference; causal estimands; bias analysis and correction; selection bias; survival and longitudinal data analysis; simulation and sensitivity analyses; missing-data methods; target trial emulation.

Applications: observational studies using linked electronic health records, registries, and cohort data; vaccine effectiveness and disease-severity evaluation; pharmacoepidemiology.

DOCTORAL RESEARCH

The University of Hong Kong *School of Public Health* 2022–Present

COVID-19 severity pyramid and in-hospital progression

Asked how changing admission practices altered COVID-19 severity estimates and longitudinal in-hospital progression. Analysed linked records for about 2.8 million cases and 113,000 hospitalisations using Bayesian severity-pyramid, Aalen–Johansen competing-risk, and Cox models.

Hospital pressure, admission selection, and fatality risk

Assessed how hospital pressure, admission selection, and admission severity shaped 30-day fatality. Analysed linked records for about 117,000 hospitalisations using wave-specific generalised additive models and model-based counterfactual standardisation.

Causal estimands and selection bias in test-negative designs

Examined when vaccine-induced severity attenuation drives test-negative estimates away from source-population causal estimands. Used DAG-informed simulation and analytic selection-bias derivations to show that severity adjustment can amplify bias whereas inverse-probability weighting can reduce it.

Hospital-based TND estimates of vaccine effectiveness *(ongoing)*

Uses a shared source-population simulation to compare trial, cohort, and hospital-based estimates and isolate selection bias from clinical eligibility, differential testing, and control-illness composition.

PUBLICATIONS/MANUSCRIPTS

Peer-Reviewed Publications

†Co-first author

1. **Liu, C.**, Okoli, G. N., Chen, R., Sullivan, S. G., and Cowling, B. J. (2026). SARS-CoV-2 vaccination and attenuation of breakthrough infection severity: A systematic global review and meta-analysis. *Clinical Infectious Diseases*, ciag346.
2. Du, Z., **Liu, C.**†, Bai, Y., Wang, L., Lim, W. W., Lau, E. H. Y., and Cowling, B. J. (2024). Predicting efficacies of fractional doses of vaccines by using neutralizing antibody levels. *JMIR Public Health and Surveillance*, 10, e49812.
3. Du, Z., **Liu, C.**†, Wang, C., Xu, L., Xu, M., Wang, L., Bai, Y., Xu, X., Lau, E. H. Y., Wu, P., and Cowling, B. J. (2022). Reproduction numbers of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) variants: A systematic review and meta-analysis. *Clinical Infectious Diseases*, 75(1), e293–e295.
4. Du, Z., Wang, C., **Liu, C.**†, Bai, Y., Pei, S., Adam, D. C., Wang, L., Wu, P., Lau, E. H. Y., and Cowling, B. J. (2022). Systematic review and meta-analyses of superspreading of SARS-CoV-2 infections. *Transboundary and Emerging Diseases*, 69(5).
5. **Liu, C.**†, Xu, L., Bai, Y., Xu, X., Lau, E. H. Y., Cowling, B. J., and Du, Z. (2022). Local surveillance of the COVID-19 outbreak. *Frontiers in Physics*, 10, 824369.

Working Papers

1. **Liu, C.**, Murphy, C., Okoli, G. N., Sullivan, S. G., and Cowling, B. J. *Vaccine-induced severity attenuation and bias in test-negative estimates of vaccine effectiveness*. Manuscript in preparation.
2. **Liu, C.**, Cheung, J. K., Lin, Y., Wu, P., Wong, J. Y., and Cowling, B. J. *Hospital pressure, admission selection, and COVID-19 hospital fatality risk across epidemic periods in Hong Kong: a retrospective cohort study*. Manuscript in preparation.

SELECTED PRESENTATIONS

EPIDEMICS^{10} — San Diego, USA
Oral presentation — *Dec 2025*
Impact of vaccine-induced severity attenuation on bias in vaccine-effectiveness estimates from test-negative designs.

EPIDEMICS^{9} — Bologna, Italy
Poster presentation — *Nov 2023*
Estimating the COVID-19 severity pyramid in Hong Kong, January 2020–January 2023.

RESEARCH EXPERIENCE

Laboratory of Data Discovery for Health (D^{2}4H) — Hong Kong SAR, China
Research Associate — *Jul 2021–Jul 2022*

- Supported data access, protocol development, and ethics approval for a Hospital Authority Data Collaboration Lab retrospective cohort study of oseltamivir effectiveness in hospitalised influenza patients using longitudinal clinical records.
- Co-developed an R package integrating EpiEstim and virosolver to simulate epidemic trajectories, estimate time-varying reproduction numbers, and support real-time monitoring of local COVID-19 transmission.

- Conducted systematic reviews and meta-analyses of SARS-CoV-2 transmissibility, synthesising variant-specific reproduction numbers and superspreading parameters.

Southern University of Science and Technology Shenzhen, China
Research Assistant, Department of Statistics and Data Science *Jul 2020–Jun 2021*

- Conducted methodological literature reviews and developed estimation methods using the MM algorithm for bivariate binomial distributions and multivariate zero-inflated count data.

TEACHING EXPERIENCE

The University of Hong Kong

CMED 6020 Advanced Statistical Methods I, *Tutor* Spring 2025–Spring 2026

CMED 6020 Advanced Statistical Methods I, *Teaching Assistant* Spring 2023–Spring 2024

Southern University of Science and Technology

MA103A Linear Algebra A, *Teaching Assistant* Fall 2019

MA102B Calculus A II, *Teaching Assistant* Spring 2019

MA101B Calculus A I, *Teaching Assistant* Fall 2018

ADDITIONAL TRAINING

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID)
University of Washington, Department of Biostatistics July 2023

Selected modules: MCMC I and II for Infectious Diseases; Simulation-Based Inference for Epidemiological Dynamics; Statistics and Modeling with Novel Data Streams.

TECHNICAL SKILLS

Programming: R, Python; **Tools:** LaTeX, Git.

LANGUAGES

Mandarin (native); English (professional); Cantonese (conversational).